Kinetic Studies on the Catalytic Ammonia Decomposition over Ni and Ru Catalysts

Paula Marie Hirte, Gerrit Küchen, Thomas Turek

Institute of Chemical and Electrochemical Process Engineering, Clausthal University of Technology, Clausthal-Zellerfeld, Germany

To achieve greenhouse gas neutrality, the share of renewable energies must be enhanced. However, their intermittency poses a major challenge and increases the need for Power-to-Gas technologies to allow for a long-term energy storage in form of chemicals, such as hydrogen. To enhance its transportability and ease of handling, hydrogen can be converted to ammonia and decomposed on site when needed [1,2]. The engineering of efficient reactors for ammonia decomposition requires a detailed understanding of the reaction mechanism and the corresponding kinetic description [3]. This study contributes to this need by means of steady-state and pulse experiments in differential fixed-beds.

For this work, Ni and Ru catalysts (10 wt.%) are produced by wet impregnation on Al_2O_3 and are physically and chemically characterized by BET, TPR, XRD, chemisorption and Hg-porosimetry. Kinetic investigations are carried out in a catalyst analyzer (BELCAT II, MicrotracBEL Corp.) and the gas composition is resolved via mass spectroscopy. Next to temperature-programmed measurements, NH_3-pulse experiments are performed at varying temperatures.

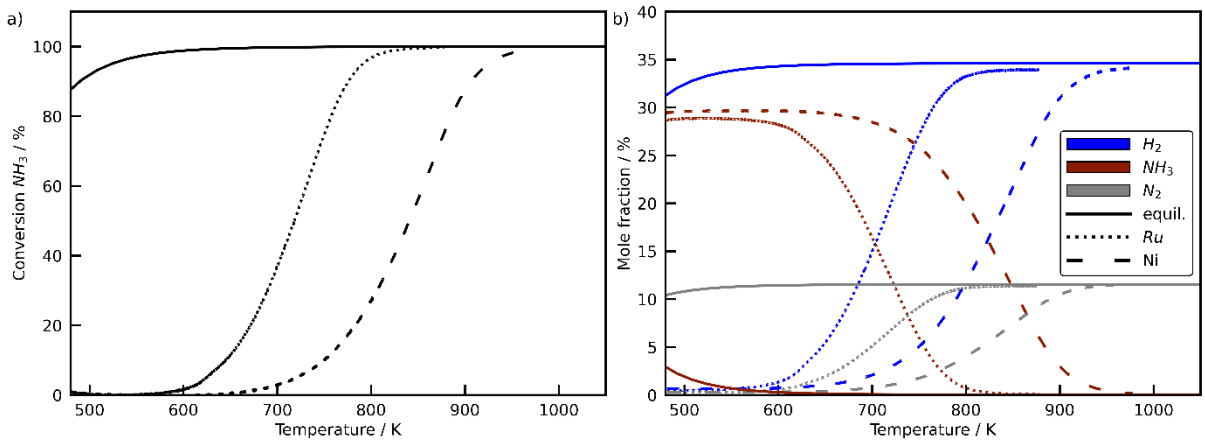

Figure 1: Temperature-programmed experiments with Ru/Al_2O_3 and Ni/Al_2O_3. a) Conversion. b) Gas composition and equilibrium. *Conditions*: 30 mg catalyst, 5 $Kmin^{-1}$, 20 $mlmin^{-1}$ 30 % NH_3 in Ar.

Figure 1 comprises the conversion and gas composition for each catalyst. The Ru catalyst reaches the thermodynamic equilibrium from 800 K onwards and is significantly more active compared to Ni catalyst over the whole temperature range.

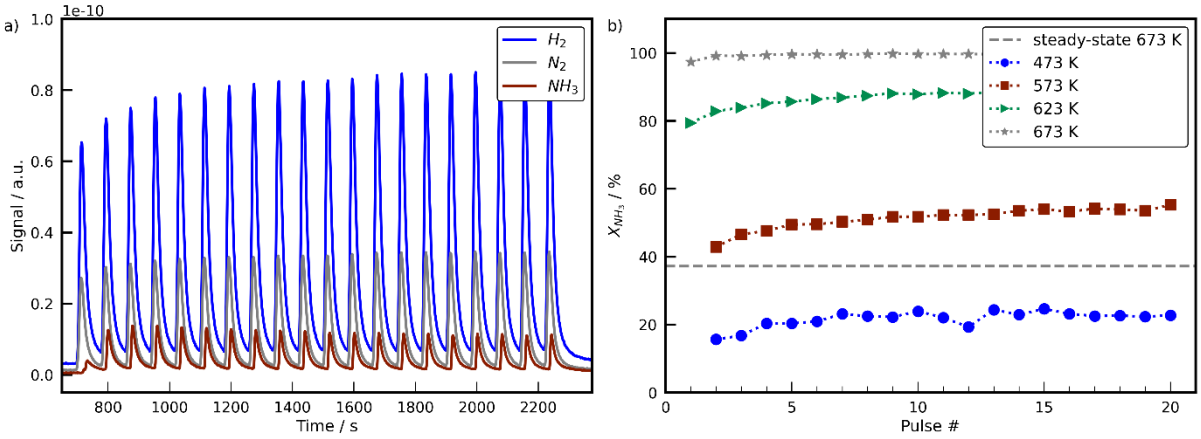

Figure 2: Results of pulsing with 30 % NH_3 in argon over 30 mg Ru/Al_2O_3. a) Overall MS signal of the pulses at 623 K. b) Calculated NH_3 conversion for each pulse at different temperatures and a comparison to a steady-state measurement at 673 K.

Figure 2a shows the raw data of a pulse experiment. To improve comparability, the ammonia conversion is calculated for every single peak. With an increasing number of pulses applied, a constant conversion is achieved indicating the involvement of adsorption phenomena. Interestingly, compared to stationary measurements, a significantly higher conversion is obtained with pulse measurements. In upcoming studies, additional dynamic measurements with different gas compositions and novel catalysts will be conducted to further investigate the kinetics of ammonia decomposition.

References

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